

MD. Tanvir Hassan

RESEARCHER IN LLM REASONING, AGENTIC SYSTEMS, AND AI SAFETY

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Research Profile

Computer Science and Engineering graduate from BUET interested in understanding how large language models reason, how efficient agentic systems can be built, and how their behavior can be interpreted and secured. Current work focuses on attack trajectories and behavioral motifs in agentic systems, along with process reward models for tabular reasoning trained using contrastive chain-of-thought steps.

Work Experience

Therap BD Ltd.

Dhaka, Bangladesh

Associate Software Engineer

Apr. 2025 – Present

- Contributed to building and maintaining a standalone authentication server; investigated vulnerabilities and applied principled security fixes.
- Maintained and improved platform-wide message queues and distributed caching to enhance scalability and reduce latency.
- Implemented passkey-based authentication and replay-attack protections; redesigned application-wide role-based access control (RBAC).

Education

Bangladesh University of Engineering and Technology (BUET)

Dhaka, Bangladesh

B.Sc. in Computer Science and Engineering

Feb. 2020 – Mar. 2025

- CGPA:** 3.96/4.00
- Notable Coursework:** Computer Security, Machine Learning, Computer Networks, Artificial Intelligence, Software Engineering, Data Structures and Algorithms, and Object-Oriented Programming.

Notre Dame College

Higher Secondary Certificate

2017 – 2019

Research Experience

TablePN: How Contrastive CoT Steps Evolve a Process Reward Model for Tables

Supervisors: Dr. Mohammed Eunus Ali, Md. Tanvir Parvez

Ongoing

- Developing process reward models for table-based tasks using contrastive chain-of-thought supervision to capture the quality of intermediate reasoning steps.
- Studying how contrastive samples and CoT structures affect reward-signal quality and downstream fine-tuning on table-reasoning tasks.

Agentic Security: Trajectory Behavioral Motif Analysis

Agentic AI Security

Ongoing

- Analyzing trajectories generated by agentic systems to identify anomalous behavioral motifs and characterize recurring attack patterns.
- Investigating whether trajectory-level behavioral patterns can generalize across attacks and agentic environments.

MapEval: A Map-Based Evaluation of Geospatial Reasoning in Foundation Models

ICML 2025 Spotlight — Supervisors: Dr. Mohammed Eunus Ali, Md. Tanvir Parvez

2025

- Evaluated foundation models on textual, API-based, and visual-map reasoning tasks spanning 180 cities in 54 countries.
- Studied navigation, spatial relationships, route planning, distance estimation, and travel planning, revealing a substantial gap between current models and human performance.

How Semantically Stable Are LLM Refusals? Measuring Confusion in Local Safety Boundaries

IEEE DSAA 2026 — Accepted, Short Presentation; Supervisor: Dr. Ch. Md. Rakin Haider

2026

- Studied inconsistent LLM refusal decisions for semantically similar benign prompts and introduced ParaGuard, a 10,000-prompt evaluation dataset.
- Developed metrics for local safety-boundary confusion to support more consistent refusal behavior and fewer false refusals.

DFCon: Attention-Driven Supervised Contrastive Learning for Robust Deepfake Detection

IEEE Signal Processing Cup 2025 — 1st Runner-Up; ICASSP 2026 Under Review; Supervisor: Dr. Mohammad Saifur Rahman

2025–2026

- Trained on eight DeepFakeBench datasets augmented with diffusion-, GAN-, and VAE-generated synthetic data.
- Fine-tuned MaxViT, CoAtNet, and EVA-02 using supervised contrastive and focal losses; built a multi-stage backbone, classifier, and majority-voting pipeline.
- Achieved 95.83% validation accuracy while addressing class imbalance.

Awards & Honors

- 2024 **Research Grant Recipient**, RISE Research Grant Award
- 2025 **1st Runner-Up**, IEEE Signal Processing Cup
- 2024 **Silver Award**, Blockchain Olympiad Bangladesh
- 2024 **2nd Runner-Up**, IUT CTF 2024
- 2023 **Champion**, DL Sprint 2.0 BUET CSE FEST 2023
- 2023 **2nd Runner-Up**, Web3 Hackathon BUET CSE FEST 2023

Technical Skills

Programming Python, C/C++, Java, JavaScript, SQL, HTML/CSS

ML & Research Large language models, chain-of-thought reasoning, process reward models, contrastive learning, model evaluation

Tools & Frameworks Linux, Shell, Git, React, Node.js, Express, Svelte/SvelteKit, PostgreSQL, Supabase

Selected Technical Projects

CTFe

Cybersecurity competition platform

- Built support for public and private CTF contests, post-contest upsolving, community chat, and technical blogs.
- Implemented solution-similarity checks to flag potentially copied submissions.
- **Technologies:** Svelte, SvelteKit, JavaScript, PostgreSQL.

FoodXpress

Food delivery and restaurant management platform

- Developed multi-restaurant ordering and real-time delivery-rider tracking.
- **Technologies:** React, Node.js, JavaScript, PostgreSQL.

HalfWall

Social platform for university students

- Built posting, groups, events, and real-time chat using socket-based communication.
- **Technologies:** EJS, Node.js, JavaScript, PostgreSQL.

Interests

Music and violin performance; traveling to new places and exploring unfamiliar cultures and ideas.